

MILESTONE 2 - TECHNICAL DOCUMENTATION

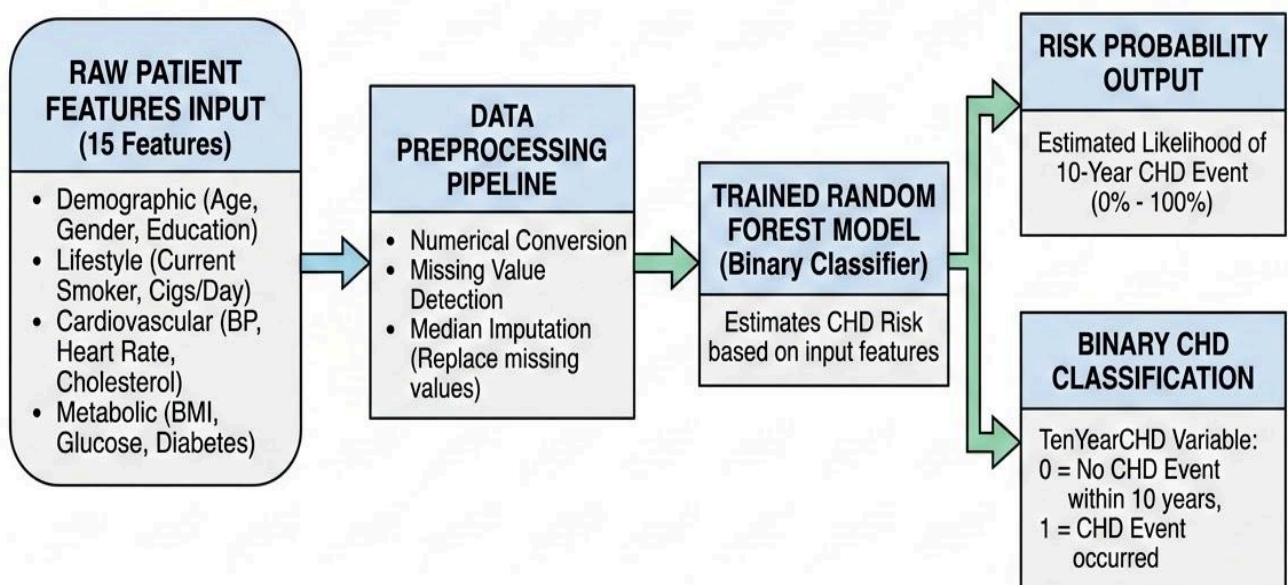
1. AI Prediction Layer – Random Forest-Based CHD Risk Prediction

1.1 Overview

The foundational component of the MediSphere platform is the **AI Prediction Layer**. Its primary objective is to evaluate the 10-year risk of coronary heart disease (CHD) utilizing clinical, demographic, lifestyle, cardiovascular, and metabolic parameters. The subsystem employs a supervised **Random Forest** classifier trained on the benchmark **Framingham Heart Study** dataset. The risk estimation task is structured as a binary classification problem where the target label, **TenYearCHD**, denotes whether a patient developed CHD within a 10-year follow-up window (where **0** indicates the absence of a CHD event and **1** denotes a diagnosed CHD event).

By mapping high-dimensional patient records into probabilistic risk profiles, this layer acts as an automated triage mechanism, eliminating manual feature analysis. The predictive model functions as an early risk-screening utility rather than a diagnostic replacement, providing research-grade risk stratified scores to assist healthcare professionals.

AI Prediction Pipeline Workflow



1.2 Dataset Description & Feature Breakdown

The underlying dataset derived from the **Framingham Heart Study** comprises 4,240 patient records. Predictive inference is based on 15 distinct clinical attributes categorised into four principal domains:

- **Demographic Features:** Age, Gender, and Education level capturing fundamental socio-demographic risk indicators.
- **Lifestyle Variables:** Current smoking status and daily cigarette consumption quantifying behavioral exposure.
- **Cardiovascular Markers:** Systolic BP, Diastolic BP, Heart Rate, Total Cholesterol, Antihypertensive Medication usage, Prevalent Hypertension, and Prevalent Stroke history evaluating arterial and cardiac strain.
- **Metabolic Indicators:** Body Mass Index (BMI), Glucose levels, and Diabetes diagnosis assessing metabolic syndrome complications.

Category	Feature Name	Clinical Relevance & Description
Demographic	Gender, Age, Education	Baseline demographic traits and social determinants influencing long-term cardiovascular baseline risk.
Lifestyle	Current Smoker, Cigarettes Per Day	Behavioral risk indicators capturing exposure to tobacco smoke and dosage per day.
Cardiovascular	BP Medication, Prevalent Stroke, Prevalent Hypertension, Diabetes, Total Cholesterol, Systolic BP, Diastolic BP, Heart Rate	Direct physiological measurements and medical history factors evaluating arterial, cardiac, and vascular stress.
Metabolic	BMI, Glucose	Metabolic markers evaluating glycemic control and body composition impacting metabolic syndrome risks.

Table 1: Clinical feature classification and domain relevance mapping for the Framingham Heart Study dataset.

1.3 Target Variable Definition

The target variable, **TenYearCHD**, is a binary indicator representing the manifestation of coronary heart disease within ten years of initial baseline assessment.

By formulating the task as binary classification, the machine learning architecture maps the 15-dimensional input vector into a calibrated probability distribution reflecting individual patient risk.

1.4 Data Preprocessing Pipeline

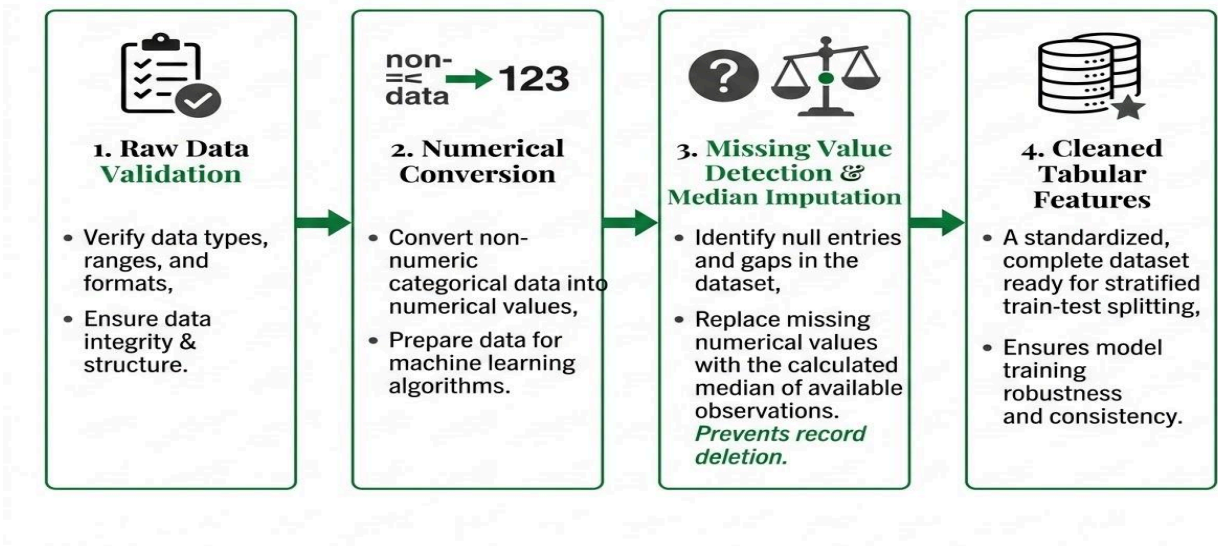
To maintain statistical integrity and prevent sample attrition, raw input vectors undergo systematic preprocessing:

- **Raw Data Validation:** Verifying data types, boundary constraints, and structural consistency across features.
- **Numerical Conversion:** Standardizing categorical and discrete inputs into numerical formats compatible with matrix operations.
- **Missing Value Detection & Median Imputation:** Identifying null values and executing median imputation. Median imputation replaces missing values with attribute-specific medians, preserving sample size without introducing parametric bias from skewed distributions.

This end-to-end preprocessing pipeline guarantees structural consistency across model training and real-time backend API inference.

Raw Patient Data » Type Validation » Categorical Conversion » Median Imputation » Cleaned Tabular Features

Data Preprocessing Workflow



1.5 Train-Test Partitioning

The preprocessed dataset is partitioned using a 80/20 stratified train-test split, allocating 3,392 samples for model training and cross-validation, while reserving 848 hold-out samples for evaluation.

Stratification ensures that the relative proportion of positive and negative CHD cases is strictly maintained across both subsets, preventing class ratio skewing caused by severe dataset imbalance.

Evaluating performance exclusively on unseen test data eliminates data leakage and ensures generalization capability.

1.6 Algorithmic Justification: Random Forest

The platform selects **Random Forest** as the core classifier. As an ensemble method aggregating decision trees via bagging, **Random Forest** provides distinct structural advantages:

- **Nonlinear Feature Interaction:** Effectively captures complex clinical interactions (e.g., compounding effects of hypertension, smoking, and age) without assuming linear boundary conditions.
- **Overfitting Mitigation:** Averages variance across randomized trees, significantly boosting generalization on tabular healthcare records.
- **Interpretable & Probabilistic Output:** Produces calibrated continuous probability estimates alongside discrete predictions, natively supporting post-hoc explainability via **SHAPTreeExplainer**.

1.7 Hyperparameter Optimization Strategy

Model tuning utilizes **GridSearchCV** with 5-fold cross-validation. Optimization optimizes the **F1 Score** rather than raw accuracy to handle class imbalance.

Hyperparameters tuned include tree count, maximum depth, and split criteria. **GridSearchCV** iteratively identifies configurations maximizing minority class detection performance.

Using the **F1 Score** ensures balanced model behavior, avoiding bias toward the majority class ("No CHD").

1.8 Inference Pipeline & REST API Integration

The optimized model is deployed via a backend REST API endpoint:

POST /api/model/predict

The endpoint accepts structured JSON feature inputs and returns predicted class labels, calculated risk probabilities, and model performance metadata for frontend rendering.

1.9 Predictive Inference Output

The model outputs discrete classification labels alongside continuous probability scores. For instance, low-risk test inferences output Class 0 probabilities of 93.5% versus Class 1 probabilities of 6.5%, providing clinicians with actionable likelihood assessments.

2. Model Analysis Layer – Statistical Performance Assessment

2.1 Overview

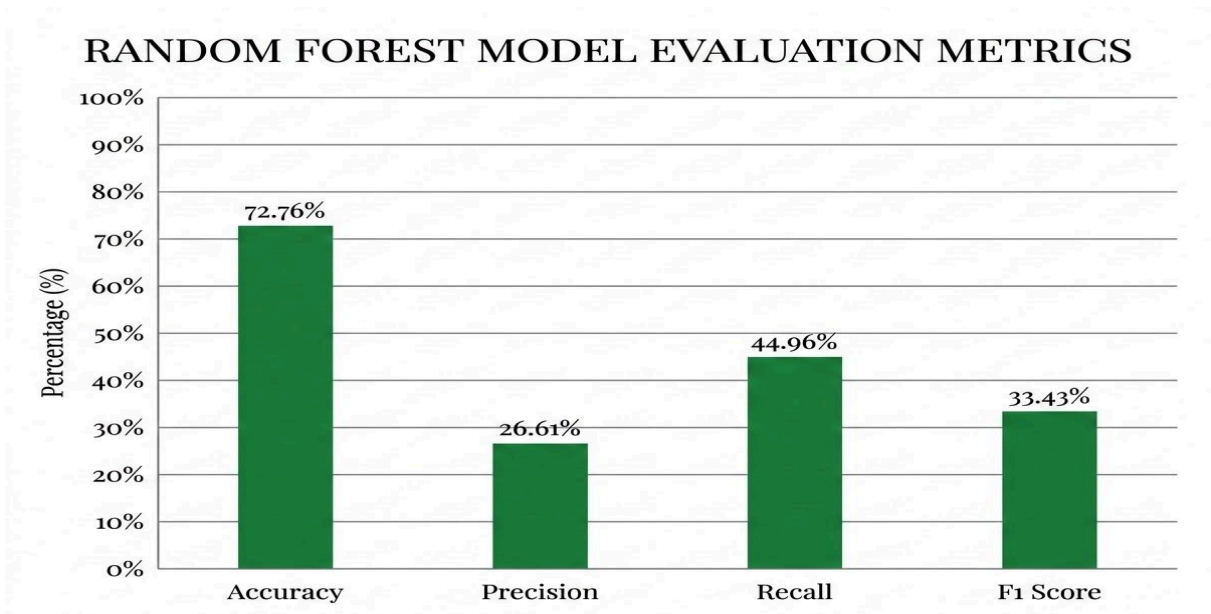
The **Model Analysis Layer** evaluates the generalizability and clinical reliability of the **Random Forest** model across multiple statistical metrics: **Accuracy**, **Precision**, **Recall**, **F1 Score**, and a full **Confusion Matrix** breakdown.

In healthcare applications, relying solely on accuracy can be misleading due to class imbalance; thorough multi-metric evaluation is essential to assess false negative risks.

2.2 Quantitative Evaluation Metrics

The model achieved an overall test accuracy of 72.76%. While accuracy measures total correct classifications across all test cases, it provides an incomplete picture in imbalanced medical contexts where detecting high-risk cases is critical.

Below is the complete statistical evaluation breakdown based on the 848 held-out test records:



Metric	Tested Value	Clinical Significance & Formula Context
Accuracy	72.76%	Overall proportion of correct predictions across all test cases: (TP + TN) / Total.
Precision	26.61%	Fraction of positive predictions that are true positive cases: TP / (TP + FP).

Metric	Tested Value	Clinical Significance & Formula Context
Recall	44.96%	Proportion of actual high-risk patients correctly identified: $TP / (TP + FN)$.
F1 Score	33.43%	Harmonic mean balancing Precision and Recall: $2 * (Precision * Recall) / (Precision + Recall)$.

2.3 Confusion Matrix Analysis

Table 2 summarizes model performance on the 848 held-out test instances:

Table 2: Random Forest evaluation metrics on test dataset.

Actual / Predicted	Predicted Low Risk (o)	Predicted High Risk (1)
Actual Low Risk (o)	559 (True Negative)	160 (False Positive)
Actual High Risk (1)	71 (False Negative)	58 (True Positive)

Table 3: Confusion matrix breakdown on hold-out test set (N = 848).

Matrix Breakdown & Classification Mechanics:

- **True Negatives (TN = 559):** Correctly identified low-risk cases.
- **False Positives (FP = 160):** Low-risk cases misclassified as high-risk, leading to unnecessary follow-up screening.
- **False Negatives (FN = 71):** High-risk cases misclassified as low-risk, representing potential missed interventions.
- **True Positives (TP = 58):** Correctly identified high-risk patients requiring early intervention.

2.4 Multi-Metric Assessment Rationale

Evaluating clinical AI models requires multi-metric analysis. While 72.76% accuracy appears acceptable, a 44.96% **Recall** rate reveals that high-risk cases remain challenging to capture fully, and a 26.61% **Precision** indicates frequent false alarms.

MediSphere presents a balanced metric profile to provide a complete evaluation of model strengths and limitations.

This transparent evaluation aligns with clinical reporting standards where accuracy alone is insufficient due to dataset class imbalance.

2.5 Model Analysis as a Validation Layer

Model analysis acts as a rigorous validation gate within the architecture:

- **Prediction Generation Layer:** Evaluates whether the model produces stable probabilistic predictions for incoming patient feature vectors.
- **Statistical Validation Layer:** Evaluates how reliably predictions perform across population-level distributions.

This validation structure prevents black-box deployment by continuously recording performance data in the backend database.

Separating inference generation from statistical validation ensures rigorous, objective model monitoring in production environments.

This transforms model assessment into an evidence-based clinical evaluation workflow.

3. Explainable AI Layer – SHAP Framework

3.1 Overview

The Explainable AI (XAI) layer integrates **SHAP** (SHapley Additive exPlanations) to address a fundamental question in clinical deployment: "What specific feature contributions drove this individual prediction?"

Using Shapley value decomposition, this subsystem provides human-interpretable feature contribution breakdowns for every generated prediction.

This feature contribution analysis bridges the gap between raw statistical probability and clinical interpretability.

In medical settings, providing a static "HIGH RISK" label without feature attribution offers insufficient context for clinical decision-making.

HIGH RISK (Unexplained Output)

Pairing predictions with Shapley attribution values provides clinicians with transparent, feature-level insights into model decisions.

3.2 The Imperative for Model Interpretability

Although ensemble models capture complex nonlinear relationships, their decision boundaries are opaque. Post-hoc explainability adds an essential translation layer between mathematical model outputs and clinical domain knowledge.

System Architecture Flow:

3.3 Mathematical Formulation: SHAP Framework

SHAP leverages game-theoretic Shapley values to calculate additive feature contributions. For tree ensembles, MediSphere implements **TreeExplainer**, an optimized algorithm that computes exact Shapley values efficiently.

Shapley attribution assigns a signed numerical value to each input feature, measuring its marginal contribution to the delta between expected baseline prediction and final model output.

Example attribution dynamics:

Elevated systolic blood pressure increases the estimated CHD risk probability, yielding a positive attribution score, whereas optimal cholesterol levels reduce the predicted risk probability, yielding a negative attribution score.

3.4 Directional Influence of SHAP Values

Directional attribution principles:

- **Positive SHAP Values (+ ϕ):** Push the model output above baseline risk, increasing the likelihood of a positive CHD classification (Class 1).
- **Negative SHAP Values (- ϕ):** Push the output below baseline risk, decreasing predicted CHD probability toward a low-risk classification (Class 0).

This bidirectional attribution framework provides far richer clinical insight than global feature importance rankings.

3.5 Explanation Generation Workflow

Execution steps for the feature-level explanation pipeline:

- **1. Feature Ingestion:** Ingest 15 cleaned patient features via REST payload.
- **2. Ensemble Inference:** Pass features through the **Random Forest** classifier to produce class probability distribution.
- **3. Shapley Calculation:** Execute **TreeExplainer** decomposition across decision paths to derive exact feature attribution values.
- **4. Payload Assembly:** Package attributions into structured JSON objects for visual rendering on the dashboard and audit persistence in PostgreSQL.

This automated pipeline makes model decisions fully transparent and auditable.

Explainability (SHAP) Calculation Pipeline

3.6 Case Study: Low-Risk Inference Attribution

In a sample inference yielding a Low Risk classification, the model calculated a 93.5% probability for Class 0 (No CHD) versus 6.5% for Class 1 (CHD Event).

Attribution analysis revealed negative **SHAP** values across top features (e.g., non-smoker status, normal systolic blood pressure, young age), confirming that these factors reduced the estimated risk score.

This demonstrates how negative attributions lower predicted probability relative to baseline risk, resulting in a Low Risk output.

Key elements unified in this explanation payload:

- **Continuous Probabilities:** Explicit likelihood scores (93.5% Class 0 vs 6.5% Class 1).
- **Discrete Risk Classification:** Final categorical label (Low Risk).
- **Quantitative Feature Attributions:** Individual Shapley values explaining the directional impact of key clinical factors.

3.7 Interpretability vs. Medical Accuracy

Integrating **SHAP** improves model transparency by revealing the mathematical factors behind predictions, converting black-box algorithms into auditable systems.

Instead of providing isolated risk labels, the system presents complete, feature-level attribution profiles.

Attribution values show how specific clinical inputs shift predicted risk relative to baseline cohort statistics.

This transparent auditing process supports clinical verification and model performance monitoring.

Important distinction: **SHAP** explains how the model reaches a decision based on training patterns; it does not guarantee clinical accuracy or serve as a direct diagnostic confirmation.

3.8 Explainability Data Persistence

Attribution outputs are persisted in PostgreSQL to maintain auditability. Stored entities include:

- **SHAP Prediction Logs:** Patient-specific prediction identifiers linked to baseline probability scores.
- **Feature Attribution Records:** Exact numerical Shapley values for all 15 clinical parameters per inference request.

Persisting attributions ensures model predictions remain fully transparent and traceable over time.

Persistence Architecture Flow:

Inference Request » TreeExplainer » JSON Payload » PostgreSQL Storage » Dashboard Audit Trail

4. Federated Learning Layer – Privacy-Preserving Collaborative Training

4.1 Overview

The **Federated Learning** layer enables decentralized, multi-institutional model training without centralizing patient datasets.

By training models locally on client nodes and aggregating parameters globally, the platform provides collaborative model optimization while upholding data privacy standards.

Multiple simulated client nodes perform local training iterations; a central orchestration server applies **Federated Averaging** to aggregate updates, using convergence detection to monitor stability.

4.2 Motivation for Federated Architecture

Centralizing patient data creates significant privacy risks and regulatory compliance challenges under standards like HIPAA and GDPR. Federated learning resolves these limitations by bringing model training directly to local data owners.

Architectural Comparison:

- **Centralized Machine Learning:** Requires raw data aggregation onto a central server, introducing privacy risks and single points of failure.
- **Federated Learning:** Keeps patient data on local client nodes. Only model updates (weights or gradients) are transmitted to the aggregation server, maintaining data privacy.

Federated vs. Centralized Learning Comparison

4.3 Client Simulation & Data Partitioning

The prototype simulates multiple independent client nodes (e.g., distinct clinical sites) using partitioned subsets of the Framingham dataset.

Partitioning methods preserve real-world non-IID (non-independently and identically distributed) data traits across client nodes.

Key simulation parameters:

- **Local Node Isolation:** Strict boundary enforcement preventing cross-node data leakage during training.
- **Parameter Transfer Protocol:** Transmitting model updates via secure API endpoints for server-side aggregation.

4.4 Local Model Training Mechanics

During each federated round, client nodes initialize local models using the latest global parameters, train on local data for set epoch thresholds, and generate updated parameter vectors.

Workflow:

Receive Global Model » Execute Local Training » Compute Parameter Delta » Upload Update Payload

Local node data remains strictly isolated throughout training; only mathematical parameter updates are transmitted externally.

4.5 Federated Averaging (FedAvg) Protocol

The platform implements the standard **Federated Averaging (FedAvg)** algorithm to aggregate client update vectors into a unified global model.

The central server aggregates parameters proportional to each client's training sample size, computing a weighted average across all participating nodes.

Aggregation Mechanics:

Weighted client parameters update the global model, which is then redistributed across client nodes to begin the next iterative training cycle.

4.6 Iterative Training Lifecycle

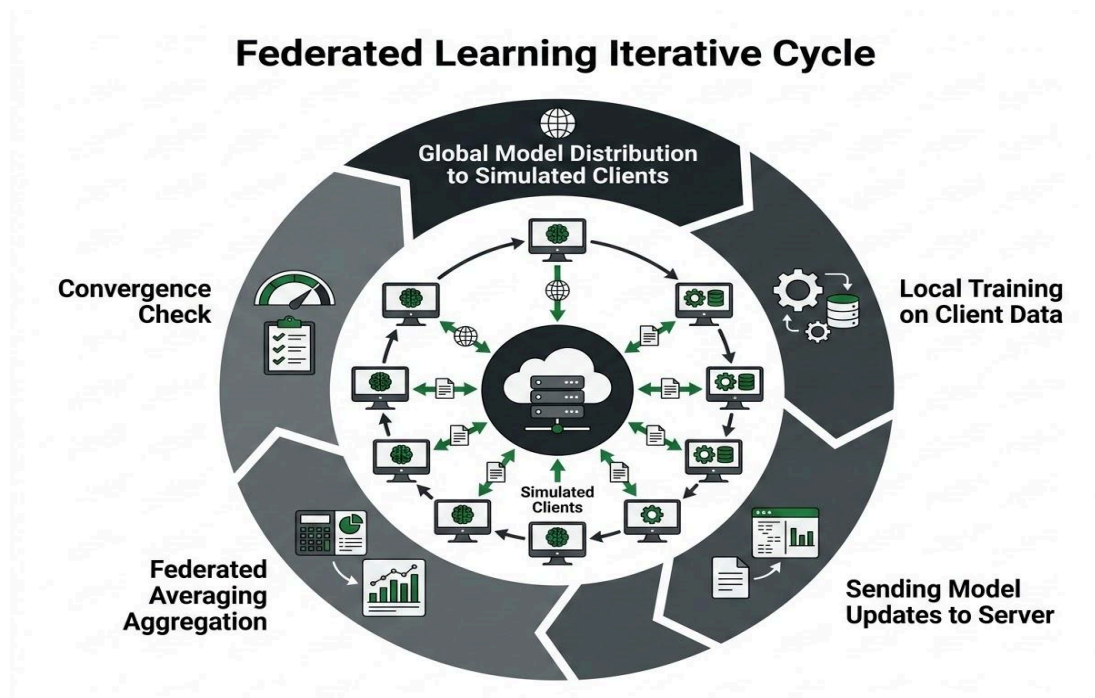
The **Federated Learning** architecture executes structured iterative cycles:

- **1. Model Broadcast:** Distributing current global weights to selected client nodes.
- **2. Local Computation:** Executing gradient descent or tree structure updates on isolated client datasets.
- **3. Update Transmission:** Sending local parameter updates to the central server via REST endpoint payloads.
- **4. Global Aggregation:** Combining client parameter updates using **FedAvg**.
- **5. Convergence Evaluation:** Evaluating global loss stabilization to check convergence criteria.

This iterative cycle continues until global model performance satisfies convergence thresholds.

Convergence detection ensures efficient resource utilization by preventing unnecessary communication rounds once model parameters stabilize.

Federated Learning Iterative Cycle



4.7 Convergence Criteria & Optimization

Convergence monitoring checks parameter delta thresholds across consecutive rounds to determine optimal training termination.

Evaluation Logic:

When global validation loss reaches a plateau and parameter updates fall below target tolerances, training halts automatically to avoid overfitting and unnecessary compute costs.

Automated convergence checking streamlines the training process while preserving optimal final model performance.

4.8 Tracking & Logging Federated Metrics

System persistence logs detailed metrics for each training round to maintain visibility into federated optimization:

- **Round-Level Metrics:** Global loss trajectories, aggregation timestamps, and client participation rates stored per iteration.
- **Convergence Diagnostics:** Parameter delta magnitudes tracking model stability over time.

These detailed logs allow operators to monitor training performance and verify node synchronization across training rounds.

Tracking these metrics ensures complete system auditability across the entire federated learning lifecycle.

Detailed performance logs provide complete visibility into model optimization and convergence behavior across rounds.

4.9 Privacy Guarantees & Security Considerations

Decentralized training significantly enhances privacy by ensuring raw patient data never leaves local client nodes, fulfilling core data protection requirements.

This architecture supports privacy-preserving machine learning across distributed medical institutions.

While decentralized training removes the need to centralize patient records, production deployments should supplement it with additional privacy-enhancing technologies:

- **Differential Privacy (DP):** Adding calibrated noise to parameter updates to prevent gradient inversion attacks.
- **Secure Aggregation Protocols:** Encrypting update transmissions so the central server can only inspect aggregated results rather than individual client payloads.

Combining federated learning with advanced privacy safeguards ensures robust, enterprise-grade data protection.

5. Subsystem Integration & Operational Flow

The core innovation of the MediSphere platform lies in the seamless integration of its four architectural layers into a unified clinical workflow.

Subsystem Interdependencies:

- **Layer 1 (AI Prediction):** Generates patient-level classification scores and continuous probability distributions.
- **Layer 2 (Model Analysis):** Evaluates population-level performance, providing validation metrics (Precision, Recall, F1 Score).
- **Layer 3 (Explainable AI):** Computes feature-level Shapley attributions explaining individual prediction outputs.
- **Layer 4 (Federated Learning):** Enables collaborative model training across decentralized nodes, continuously improving global weights without compromising privacy.

This modular architecture creates a complete, privacy-aware framework for clinical risk prediction.

6. Integration: Prediction & Explainability

The AI Prediction Layer and Explainable AI Layer operate in a tight inference-attribution loop.

Step 1: Predictive Inference

The **Random Forest** classifier evaluates patient features and outputs a calibrated risk probability vector.

Step 2: Post-Hoc Attribution

The **SHAPTreeExplainer** decomposes the output probability across active decision trees to extract feature attributions.

Executing prediction and attribution in a single pipeline ensures every generated output includes complete feature-level explanations.

Combining prediction with feature attributions gives clinicians the transparent rationale needed to evaluate model risk outputs.

7. Integration: Model Analysis & Explainability

Model Analysis and Explainable AI operate at complementary structural levels:

- **Model Analysis (Macro-Level Evaluation):** Evaluates population-level performance, measuring overall accuracy, precision, recall, and error distributions across cohort datasets.
- **Explainable AI (Micro-Level Attribution):** Decomposes specific individual predictions to isolate feature-level contributions driving individual risk outputs.

Together, macro-level evaluation and micro-level attributions provide a complete framework for clinical model assessment.

Combining population-level validation with individual attribution delivers both statistical rigor and clinical interpretability.

Structural Complementarity:

While macro metrics validate overall model accuracy across patient populations, local **SHAP** attributions explain individual predictions to ensure clinical transparency.

This dual approach ensures both population-level statistical validity and patient-level clinical interpretability.

8. Integration: Federated Learning & Random Forest

Combining decision-tree ensembles with federated aggregation requires structured parameter synchronization across client nodes.

Rather than attempting to average non-differentiable decision tree topologies directly, the system uses parameter aggregation protocols tailored for tree structures.

Supported aggregation approaches:

- **Ensemble Tree Union Pooling:** Aggregating decision trees from participating client models into an expanded global ensemble forest.
- **Histogram & Gradient Averaging:** Aggregating feature split histograms across client nodes to guide global tree construction.

9. Data Persistence Architecture

MediSphere relies on PostgreSQL for persistent data storage, logging model states, inference outputs, and federated training metrics across four core relational schema domains:

Database Entity	Persistence Layer	Stored Information & Scope
ML Model & Metrics	PostgreSQL	Model metadata, hyperparameter settings, accuracy, precision, recall, F1 score.
Model Predictions	PostgreSQL	Patient feature inputs, predicted class label, estimated class probabilities.
SHAP Predictions & Features	PostgreSQL	Feature-level Shapley contribution values, validation records, explanation logs.
Federated Rounds & Metrics	PostgreSQL	Training round logs, client update summaries, convergence states, global performance history.

10. Complete Technical Architecture

The overall system architecture integrates prediction, explainability, federated learning, and persistent storage into an enterprise-grade web application platform.

Core architectural components:

- **User / Client Dashboard:** A interactive web interface for patient feature input, risk score visualization, and SHAP attribution plots.
- **Backend REST Gateway Router:** An API service that orchestrates payload routing across prediction, explainability, and federated learning modules.
- **Four-Layer AI Subsystem Engine:** Executes real-time inference, computes SHAP attributions, processes evaluation metrics, and coordinates federated updates.
- **PostgreSQL Data Persistence Domain:** Central database logging all model states, inference outputs, feature attributions, and federated training metrics.

11. Final Deep-Dive Summary of the Four Components

Component 1 — AI Prediction Layer

The AI Prediction Layer uses a **Random Forest** classifier trained on 4,240 records from the **Framingham Heart Study** dataset to predict 10-year CHD risk. Preprocessing handles missing values using median imputation and partitions data using an 80/20 stratified split. Model tuning relies on **GridSearchCV** optimizing the **F1 Score** via 5-fold cross-validation. Inferences are served via a REST API endpoint (</api/model/predict>) returning risk scores and classification labels.

Component 2 — Model Analysis Layer

The Model Analysis Layer evaluates classifier performance across multiple statistical metrics: 72.76% **Accuracy**, 26.61% **Precision**, 44.96% **Recall**, and a 33.43% **F1 Score**, verified using a full **Confusion Matrix**. Multi-metric evaluation provides an essential safeguard in imbalanced clinical domains where relying solely on accuracy can hide high-risk misclassifications.

Component 3 — Explainable AI Layer (SHAP)

The Explainable AI Layer integrates **SHAPTreeExplainer** to calculate exact feature attributions for every **Random Forest** output. Signed Shapley values show whether individual clinical features increase ($+\phi$) or decrease ($-\phi$) predicted risk relative to baseline cohort statistics. All attributions are saved to PostgreSQL for auditability and visualized on the dashboard to ensure model interpretability.

Component 4 — Federated Learning Layer

The Federated Learning Layer enables privacy-preserving collaborative training across decentralized client nodes. Client nodes execute local model training on isolated data subsets and send parameter updates to the central server, which aggregates updates using **Federated Averaging (FedAvg)**. Convergence tracking monitors parameter loss across rounds to optimize training efficiency while keeping raw patient data secure.

12. Executive Architectural Summary

MediSphere integrates four core technical layers into a unified, privacy-aware clinical prediction platform. Predictive inference uses a **Random Forest** classifier trained on the benchmark **Framingham Heart Study** dataset. Model evaluation combines **Accuracy**, **Precision**, **Recall**, **F1 Score**, and confusion matrix analysis to ensure statistical rigor in imbalanced clinical domains. Post-hoc explainability powered by **SHAPTreeExplainer** provides individual, feature-level attributions for transparent decision support. Finally, decentralized **Federated Learning** aggregated via **Federated Averaging** enables privacy-preserving collaborative model optimization across distributed institutions.

Four-Layer System Architectural Overview:

- **Layer 1 (AI Prediction):** Predictive inference via optimized **Random Forest** classifier trained on the **Framingham Heart Study** dataset.
- **Layer 2 (Model Analysis):** Multi-metric validation assessing Precision, Recall, F1 Score, and Confusion Matrix metrics.
- **Layer 3 (Explainable AI):** Feature-level attribution using **SHAPTreeExplainer** for transparent clinical decision support.
- **Layer 4 (Federated Learning):** Privacy-preserving collaborative training aggregated via **Federated Averaging (FedAvg)**.

End-to-End System Pipeline

